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The microbiome in graft-versus-host disease: a tale of two ecosystems

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Abstract
Graft-versus-host disease (GVHD), a life-threatening complication of allogeneic hematopoietic stem cell transplantation (HSCT), is shaped by a dynamic interplay between two microbial ecosystems: the recipient's disrupted microbiome and the donor's transplanted microbiota. This narrative review unravels the "tale of two ecosystems," exploring how pre-transplant chemotherapy, radiation, and antibiotics induce recipient dysbiosis—marked by loss of beneficial taxa (*Clostridia*, *Faecalibacterium*) and dominance of pathobionts (*Enterococcus*). These shifts impair barrier integrity, fuel systemic inflammation, and skew immune responses toward pro-inflammatory T-cell subsets, exacerbating GVHD. Conversely, emerging evidence implicates donor microbiota in modulating post-transplant immune reconstitution, though its role remains underexplored. Therapeutic strategies, including probiotics, prebiotics, and fecal microbiota transplantation (FMT), demonstrate promise in restoring microbial balance, enhancing short-chain fatty acid (SCFA)-driven immune regulation, and reducing GVHD severity. However, challenges such as strain-specific efficacy, safety in immunocompromised hosts, and protocol standardization persist. By bridging microbial ecology and immunology, this review underscores the microbiome's transformative potential in redefining GVHD management and advocates for personalized, microbiome-targeted interventions to improve HSCT outcomes.

Keywords GVHD, Graft-versus-host disease, Allogeneic hematopoietic stem cell transplantation, Microbiome, Immune response, Short-chain fatty acid

Introduction

Graft-versus-host disease (GVHD) remains a formidable challenge in allogeneic hematopoietic stem cell transplantation (HSCT), contributing significantly to morbidity and mortality [1]. This complex immunological disorder arises when donor immune cells attack recipient tissues, with the gastrointestinal tract being a primary target. Recent advances have illuminated the pivotal role of the gut microbiome in modulating immune responses and shaping GVHD outcomes [2, 3]. The dynamic interplay between the recipient's disrupted microbial ecosystem—altered by pre-transplant therapies such as chemotherapy, radiation, and antibiotics—and the donor's transplanted microbiota introduces a novel dimension to GVHD pathogenesis [4, 5]. Dysbiosis,

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characterized by the loss of beneficial microbial taxa and the overgrowth of pathobionts, disrupts immune homeostasis, impairs barrier function, and amplifies pro-inflammatory responses, thereby exacerbating GVHD severity [6]. Conversely, the donor microbiome, introduced via the graft, holds potential to influence post-transplant immune reconstitution and disease progression [7]. This narrative review explores the "tale of two ecosystems," delving into the mechanisms by which recipient and donor microbiomes converge to influence GVHD. By examining the immunological and microbial crosstalk, we highlight emerging therapeutic strategies—such as probiotics, prebiotics, and fecal microbiota transplantation—that aim to restore microbial balance and mitigate GVHD, paving the way for personalized, microbiome-targeted interventions to enhance HSCT success.

The recipient microbiome: A disrupted ecosystem

Pre-transplant disruptions

The recipient's microbiome undergoes significant alterations well before hematopoietic stem cell transplantation (HSCT), driven by multiple factors such as underlying hematologic malignancies, prior treatments including chemotherapy and radiation, and pre-transplant conditioning regimens. These disruptions collectively foster a dysbiotic state with profound implications for transplant outcomes and the development of graft-versus-host disease (GVHD) (Table 1). Patients undergoing HSCT, often diagnosed with conditions like leukemia, lymphoma, or myelodysplastic syndromes (MDS), exhibit baseline dysbiosis prior to therapeutic interventions, characterized by reduced microbial diversity and an enrichment of pathobionts. For instance, leukemic cells secrete inflammatory cytokines such as tumor necrosis factor-alpha (TNF-α) and interleukin-6 (IL-6), which compromise epithelial integrity and promote the proliferation of opportunistic pathogens like *Enterococcus* and *Escherichia* [8]. Additionally, the recipient's microbial diversity declines further due to repeated cycles of remission-induction chemotherapy, often combined with total body irradiation (TBI), aimed at reducing tumor burden and suppressing the recipient's adaptive immune system to prevent graft rejection [9]. Chemotherapeutic agents, while targeting malignant cells, also damage intestinal epithelial cells, disrupting the gut barrier and facilitating microbial translocation into the systemic circulation. Montassier et al. (2015) demonstrated that patients with non-Hodgkin's lymphoma (NHL) receiving high-dose chemotherapy experienced a 70% reduction in *Firmicutes* and *Actinobacteria* alongside a 15-fold increase in *Proteobacteria* [10]linked to impaired nucleotide metabolism and compromised mucosal repair pathways, heightening susceptibility to bacterial translocation [11]. Similarly, radiation therapy exacerbates mucosal injury, diminishing Clostridiales populations essential for maintaining anaerobic conditions and suppressing pro-inflammatory γ-*Proteobacteria* [8, 12].

Myeloablative conditioning regimens, including antibiotics and antifungals, inflict collateral damage on commensal microbes. Broad-spectrum antibiotics such as fluoroquinolones deplete *Blautia* and *Faecalibacterium*—genera essential for producing short-chain fatty acids (SCFAs) like butyrate, which regulate regulatory T (Treg) cell differentiation and epithelial homeostasis [2]. The loss of obligate anaerobes creates an ecological niche for pathogenic organisms, such as *Enterococcus* species, which are implicated in GVHD development and increased post-transplant mortality [13]. Prophylactic and empiric broad-spectrum antibiotics, commonly administered to prevent systemic infections or manage neutropenic fever, exacerbate this dysbiosis. A retrospective study reported that early use of broad-spectrum antibiotics between days −7 and 0 prior to allogeneic HSCT reduced *Clostridiales* abundance and was associated with elevated transplant-related mortality [14]. Antifungal prophylaxis with azoles selectively suppresses

Table 1 Key factors in pre-transplant disruptions of recipient microbiota

Disruptive Factor	Key Examples	Mechanisms	Microbial Impact	Ref.
Underlying Malignancies	Leukemia; lymphoma, MDS	Inflammatory cytokine secretion (TNF-α, IL-6); Disruption of epithelial integrity	Reduced diversity; Enrichment of opportunistic pathobionts (Enterococcus, Escherichia)	[8]
Chemotherapy	HDC in NHL patients	Mucosal injury; Impaired nucleotide metabolism; Bacterial translocation	70% decrease in Firmicutes and Actinobacteria; 15-fold increase in Proteobacteria	[10, 11]
Radiation Therapy	TBI in allo-HCT recipients	Amplified mucosal injury; Disruption of anaerobic conditions; Expansion of pro-inflammatory γ-proteobacteria	Reduction in Clostridiales	[8, 12]
Antibiotics	Fluoroquinolones, β-lactams	Reduced SCFA production (butyrate); Epithelial barrier dysfunction	Loss of Blautia, Faecalibacterium; Dysregulation of Treg differentiation and epithelial homeostasis; Overgrowth of resistant pathogens (Enterococcus)	[2, 13]
Antifungal Prophylaxis	Azoles	Suppression of Saccharomyces species; Impaired TH17 response	Increased risk of Candida overgrowth	[15, 16]

Abbreviations: MDS: myelodysplastic syndrome; TNF-α: tumor necrosis factor-alpha; IL-6: interleukin-6; HDC: high-dose chemotherapy; NHL: non-Hodgkin's lymphoma; TBI: total body irradiation; allo-HCT: allogeneic hematopoietic cell transplantation; SCFA: short-chain fatty acid; Treg: regulatory T cell; TH17: T helper 17

Saccharomyces species, impairing T helper 17 (TH17)-mediated defenses against *Candida* overgrowth [15]. These interventions create a "vacuum effect," enabling antibiotic-resistant pathogens like vancomycin-resistant *Enterococcus* (VRE) to dominate [16]. A meta-analysis of 1,362 HSCT recipients revealed that pre-transplant enterococcal dominance increased the risk of grade III-IV acute GVHD by 3.2-fold [17].

The clinical consequences of pre-transplant dysbiosis are multifaceted, predisposing patients to opportunistic infections and disrupting immune regulation. The loss of microbial diversity and overgrowth of pathogenic bacteria activate pattern recognition receptors (PRRs), such as Toll-like receptors (TLRs), on host immune cells, eliciting an exaggerated inflammatory response [18]. This inflammatory cascade amplifies GVHD by enhancing donor-derived T-cell activation and exacerbating tissue damage in target organs, including the skin, gut, and liver [19]. Furthermore, dysbiosis-driven microbial translocation into the bloodstream contributes to systemic inflammation, sepsis, and poorer transplant outcomes [13]. Reduced microbial diversity correlates with increased intestinal permeability, evidenced by elevated serum levels of citrulline and zonulin [20]promoting endotoxin translocation and systemic inflammation via TLR4/NF-κB signaling [21]. The depletion of SCFA-producing *Clostridia* also diminishes IL-22 production by innate lymphoid cells (ILCs), impairing mucosal wound healing [22]. Clinically, these disruptions heighten susceptibility to bloodstream infections (BSIs) and non-relapse mortality (NRM), with one study reporting a 67% increase in 1-year NRM among patients with pre-HSCT dysbiosis [20].

Insights into pre-transplant microbial disruptions highlight opportunities for therapeutic intervention. Strategies such as administering *Clostridia*-enriched probiotics or fecal microbiota transplantation (FMT) during the pre-transplant period may restore epithelial integrity and mitigate GVHD risk [23]. Similarly, narrow-spectrum antibiotics, such as fidaxomicin for *Clostridioides* difficile, could preserve commensal diversity while targeting specific pathogens [24]. Future clinical studies are essential to validate these approaches and elucidate their impact on the microbiome-immune axis in HSCT recipients.

The recipient Microbiome and immune function

The microbiome of the recipient is the main regulator of immunological homeostasis, therefore influencing both innate and adaptive responses before and during HSCT. This balance is upset by dysbiosis, which drives immune cell numbers toward pro-inflammatory phenotypes, aggravating GVHD. Notably, age-specific differences in microbiome composition further modulate immune responses, particularly in pediatric and adult HSCT recipients. Schluter et al. (2020) demonstrated that age-dependent immune profiling in HSCT recipients is influenced by the gut microbiota, with younger patients exhibiting distinct immune cell dynamics, such as higher baseline frequencies of naïve T cells, which correlate with unique microbial taxa [25]. Similarly, Galloway-Peña et al. (2020) reported age-specific microbial differences, where pediatric patients showed reduced diversity of butyrate-producing bacteria compared to adults, potentially increasing susceptibility to GVHD due to impaired Treg differentiation [9]. These age-related variations underscore the need to consider developmental differences in microbiome-immune interactions when designing interventions for GVHD.

The intestinal microbiota's role in GVHD exhibits age-specific patterns, particularly in pediatric patients undergoing allogeneic HSCT, where unique microbial and immune dynamics influence disease outcomes. Biagi et al. (2015) characterized gut microbiota trajectories in pediatric HSCT patients, revealing distinct microbial shifts post-transplant, including a decline in Faecalibacterium and an increase in Proteobacteria, which correlate with GVHD severity and prolonged immune reconstitution [26]. These findings highlight the unique susceptibility of pediatric patients to microbiome-driven GVHD, driven by developmental differences in microbial ecology and immune maturation. Tailored interventions, such as age-specific probiotic regimens or FMT, may mitigate these risks by restoring microbial diversity and promoting tolerogenic immune responses in this vulnerable population.

The complex microbiome community supports homeostasis and plays a crucial role in shaping immune responses through several vital mechanisms (Table 2). Commensal microbes directly influence innate immunity through pathogen-associated molecular patterns (PAMPs) and metabolites. For instance, *Bacteroides fragilis*-derived polysaccharide A (PSA) activates TLR2 on dendritic cells (DCs), inducing IL-10 production and regulatory macrophage differentiation [27]. Conversely, dysbiosis-driven LPS from *Escherichia* activates TLR4, promoting NLRP3 inflammasome assembly and IL-1β release, which drive neutrophil infiltration and mucosal damage [28]. Commensal microbes also metabolize dietary fibers and host mucins into SCFAs such as butyrate, propionate, and acetate [29]. These SCFAs enter the circulation and interfere with immune cell activation and pro-inflammatory cytokine production. For instance, butyrate, a keystone metabolite from *Clostridia*, enhances the phagocytic capacity of neutrophils by upregulating ROS-generating enzymes like NADPH oxidase [30]. SCFAs interact with G-protein-coupled receptors (GPCRs) located on macrophages and DCs,

Table 2 Immunoregulatory functions of the recipient's microbiota and impact on HSCT outcomes

Immunological Functions	Microbial Components	Mechanisms	Outcomes	Effect on GVHD	Ref.
Innate Immunomodulation	Polysaccharide A (B. fragilis), LPS (E. coli)	TLR2/IL-10 axis, TLR4/NLRP3/IL-1β axis	Induction of regulatory MQ differentiation; Reduced inflammatory responses	Protective	[27–31]
	Butyrate (Clostridia); Acetate; Propionate	GPCRs interaction; Upregulation of ROS-generating enzymes (NADPH oxidase)	Reduced pro-inflammatory cytokine production; Enhanced anti-inflammatory responses		
Adaptive Immunomodulation	Butyrate-producing Clostridia (IV, XIVa); F. prausnitzii	HDAC inhibition; Enhancement of tight junction proteins	Increased colonic Foxp3+ Treg expansion; Induction of immune tolerance	Protective	[36–38]
	Intestinal bacteria (Bifidobacterium, Lactobacillus)	Microbiota-derived antigen presentation; MHC-II-dependent T cell activation	Increased IgA production and mucosal immunity		
	IAld	AhRs activation on ILC3	Induction of IL-22; Increased mucosal repair and epithelial homeostasis		
	Bacteroides thetaiotaomicron (succinate)	Succinate-dependent MQ activation	Increased inflammatory responses in dysbiotic conditions	Detrimental	[44]
	Clostridium scindens	BA transformation; FXR signaling	Suppression of TH1 differentiation	Protective	[45, 46]

Abbreviations: AhR: aryl hydrocarbon receptor; BA: bile acid; DC: dendritic cell; FXR: farnesoid X receptor; GVHD: graft-versus-host disease; HDAC: histone deacetylase; HSCT: hematopoietic stem cell transplantation; IL: interleukin; ILC3: group 3 innate lymphoid cell; LPS: lipopolysaccharide; MHC-II: major histocompatibility complex class II; NHL: non-Hodgkin's lymphoma; NLRP3: NOD-, LRR- and pyrin domain-containing protein 3; PSA: polysaccharide A; SCFA: short-chain fatty acid; TH: T helper cell; TLR: toll-like receptor; Treg: regulatory T cell

reducing pro-inflammatory cytokine production while enhancing anti-inflammatory responses [31]. In HSCT, the depletion of SCFA-producing microorganisms due to dysbiosis may lead to uncontrolled inflammation, thereby facilitating the development of GVHD [32].

The microbiome signals also regulate the trafficking and function of various immune cells, including alveolar macrophages, B-cells, DCs, T-cells, and group 3 innate lymphoid cells (ILC3s), through complex bidirectional interactions, shaping adaptive immune responses [33]. Intestinal DCs represent key mediators between the microbiome and adaptive immunity. These cells sample microbial antigens and migrate to mesenteric lymph nodes, where they present these antigens to naïve T cells [34]. Microbiome-derived signals heavily influence the phenotype and function of DCs. Butyrate-producing bacteria, for example, induce tolerogenic CD103+ DCs that promote Treg differentiation through retinoic acid production [35]. Conversely, pathobionts like *Enterococcus faecium* induce inflammatory DCs that drive TH1 and TH17 differentiation, exacerbating GVHD [25]. The recipient microbiome also directly influences T cell development and function, pivotal in GVHD pathogenesis. Commensal bacteria such as *Clostridium* species, particularly those belonging to *Clostridia* clusters IV and XIVa, have been shown to promote the differentiation and expansion of colonic Tregs through the production of SCFAs [36]. These Tregs express high levels of the transcription factor Foxp3 and exhibit potent immunosuppressive functions. With a similar protective and anti-inflammatory capacity, *Faecalibacterium prausnitzii* induces TGF-β and retinoic acid in DCs, fostering Foxp3+ Treg expansion [37]. In HSCT recipients with dysbiosis, the depletion of these beneficial bacteria results in reduced Treg frequencies and functions, compromising immune tolerance mechanisms [13]. In contrast, *Enterococcus*-dominated microbiota upregulate IL-6 and IL-23, skewing CD4+ T cells toward TH17 and TH1 subsets [38]. Han et al. (2020) demonstrated that *Enterobacteriaceae* enrichment reduced the Treg/TH17 ratio by 40% in acute GVHD patients, correlating with histone H3 hyperacetylation in CD4+ T cells [39]. Similarly, murine models revealed that *Lactobacillus*-enriched microbiota exacerbate GVHD via MHC-II-dependent activation of alloreactive T cells [40]. Beyond T cells, the microbiome shapes B cell development and antibody production. Intestinal bacteria, especially those in the small intestine, promote the development of IgA-producing plasma cells through T cell-dependent and independent mechanisms. IgA translocates across the epithelium into mucus secretions that are swallowed and transported up the trachea to the lungs, providing immune protection [41]. Dysbiosis-associated reductions in IgA contribute to increased bacterial translocation and systemic inflammation during HSCT [42].

The balanced microbiome is increasingly recognized as a potential contributor to immunomodulation and a promising area for intervention, inspiring the future of immunology and transplant medicine. Upset of the balanced microbiota composition disrupts this equilibrium, skewing immune cell populations toward pro-inflammatory phenotypes that exacerbate GVHD. As mentioned in HSCT recipients, loss of variety correlates with immune dysregulation, which is marked by exaggerated inflammation and compromised tolerance [13]. For instance, the association of lower microbial diversity with increased pro-inflammatory cytokines levels and GVHD key mediators, including IL-6 and TNF-α, has been underscored [18]. Conversely, higher microbial diversity has been correlated with reduced GVHD severity and improved survival outcomes, reinforcing the importance of the recipient microbiome as the mediator of immunologic tolerance and keeper of homeostasis [18, 22]. In addition, microbial metabolites, particularly butyrate, acetate, and propionate produced by fermentation of dietary fiber by *Clostridia* species, exhibit potent immunomodulatory effects. Butyrate inhibits histone deacetylases (HDACs) in colonocytes, promoting barrier function through increased tight junction protein expression, and in immune cells, enhancing Treg differentiation while suppressing pro-inflammatory cytokine production [43]. Beyond SCFAs, tryptophan derivatives like indole-3-aldehyde (IAld) activate aryl hydrocarbon receptors (AhRs) on ILC3s, promoting epithelial homeostasis and stimulating IL-22 production for mucosal repair [8]. Conversely, certain bacterial species produce metabolites that exacerbate inflammation. For instance, *Bacteroides thetaiotaomicron* generates succinate during dysbiosis, which enhances inflammatory macrophage activation [44]. Besides, dysbiosis-associated trimethylamine N-oxide (TMAO) inhibits HDAC3 in DCs, amplifying NF-κB-driven inflammation [45]. Bile acids (BAs) transformed by *Clostridium scindens* suppress TH1 differentiation via FXR signaling, while primary BAs from *Bacteroides*-deficient microbiota promote CXCL10-mediated T-cell recruitment to the gut [46]. These findings converge that the recipient microbiome profoundly influences immune function through complex interactions with innate and adaptive immune components. Dysbiosis before and during HSCT disrupts these interactions, creating a pro-inflammatory environment conducive to GVHD development.

The donor microbiome: A new contender

Traditionally, the focus has been on the immunological compatibility between donor and recipient to mitigate risks such as GVHD. However, the introduction of donor microbiota during allogeneic HCT (alloHCT) has led to a paradigm shift in understanding GVHD pathophysiology. Emerging data points to a possible major influence on post-transplant outcomes of the donor's microbiota introduced during transplantation [24]. Although the transplant procedure itself involves the infusion of donor HSCs and immune cells, new research indicates that microbial components linked with the donor might affect post-transplant results [47]. Chemotherapy and total body irradiation, among other conditioning treatments, drastically alter the recipient's microbiota, producing an ecological void that helps donor-derived bacteria colonization [24]. Broad-spectrum antibiotics aggravate this dysbiosis by depleting commensal bacteria and lowering colonizing resistance, allowing donor microbiota to spread in the gastrointestinal tract (GI) [5].

The arrival of the donor microbiota

The primary source of donor microbiota introduction is through the hematopoietic graft itself. HSC grafts, whether harvested from bone marrow (BM), peripheral blood stem cells (PBSCs), or umbilical cord (UC) blood, are not sterile and contain microbial components reflective of the donor's microbiome. Recent studies have highlighted the presence of microbial DNA and viable microorganisms in stem cell products, which are introduced into the recipient alongside immune cells and other cellular components [24]. For example, recent investigations demonstrated that bacterial taxa such as *Enterococcus* and *Staphylococcus* can be detected in grafts prepared for allo-HSCT [48]. These microbial entities may derive from the donor's microbiota or from contamination during the harvest or processing of the graft. Upon transplantation, these microbes enter a recipient environment characterized by immunosuppression, damaged mucosal barriers, and extensive microbial depletion, allowing donor-associated bacteria the opportunity to establish themselves within the host [49]. The arrival of donor microbiota is not limited to FMT. Emerging data suggest that donor immune cells, preconditioned by their native microbiota, may indirectly transfer microbial metabolites or microbial-associated molecular patterns (MAMPs) during transplantation. One example is the potential for DCs derived from donors with diverse microbiomes to induce regulatory phenotypic in recipient T cells, followed by decreasing inflammatory responses [50]. These avenues for donor microbiota introduction emphasize the possibility of donor microbiota's influences on recipient immune responses and GVHD development. Nevertheless, the extent to which donor microbiota persists and integrates into the recipient's microbiome remains an area of active investigation.

The battle for dominance

The competitive interactions between recipient and donor bacteria occur inside a host environment, which experiences fast immunological and bacterial alterations. Conditioning regimens and antibiotics diminish colonization resistance, a phenomenon where resident microbes inhibit pathogen overgrowth, thereby creating an environment conducive to donor microbiota establishment [12]. Introduced via the graft or intimate donor-receiver contact, donor-derived bacteria may use this ecological vacuum to settle into the recipient's gut. However, this process is not passive; donor and recipient microbes compete for nutrients and ecological niches (Table 3). Studies have revealed that specific bacterial taxa, like *Enterococcus*, often at the expense of helpful commensals like *Lactobacillus* and *Bifidobacterium*, can dominate the gut microbiota of allo-HSCT patients [51]. Residual recipient *Enterococcus* may exploit the post-transplant niche to dominate the gut, exacerbating mucosal damage and systemic inflammation [50]. In contrast, *Clostridiales* species from donors can modulate the recipient's immune system through various mechanisms, including producing SCFAs like butyrate, which reinforce intestinal epithelial integrity and suppress pro-inflammatory T cell responses [5]. Linked to GVHD severity, this microbial imbalance suggests that the competing dynamics between donor and recipient microbiota can directly affect disease outcomes, highlighting the therapeutic potential of modulating microbial composition to favor beneficial metabolites. Interventions such as prebiotics or synbiotics that favor donor microbiota colonization could tilt the balance toward a tolerogenic microbiome. Additionally, targeting microbial metabolic pathways, such as SCFA production, might synergize with immunosuppressive therapies to reduce GVHD severity.

On the other hand, the potential for donor-derived microbes to influence recipient immune responses extends beyond direct microbial interactions. Zhang et al. (2022) demonstrated that donor microbiota regulates erythrophagocytosis by BM macrophages, influencing local iron levels critical for HSC differentiation [52]. Microbiota-depleted mice exhibited impaired iron recycling, which skewed HSC fate toward self-renewal rather than differentiation, potentially delaying immune reconstitution [52]. This finding suggests that donor microbiota compete with resident microbes and systemically influence hematopoietic recovery.

The Microbiome-GVHD Axis

Early Immune-Microbiome interactions

The interaction between the immune system and microbes, including the microbiome, dates back to the prenatal period. Although the existence of a placental microbiome remains a contentious issue—with studies often yielding conflicting results—metabolites and antigens derived from maternal microbes have been shown to traverse the placenta, thereby influencing immune system development in both humans and mice [53, 54]. In murine models, these metabolites include AhR ligands, retinoic acid, and fatty acids. Notably, AhR ligands and retinoic acid stimulate the development of ILC3 and lymphoid tissue inducer cells, both of which play pivotal roles in establishing mucosal barrier [55, 56].

At birth, the neonate's alternative programed immune system is exposed to a vast array of microbes, yet this encounter does not precipitate harmful inflammation or exaggerated immune responses; rather, it reprograms the immune system in preparation for postnatal life [57]. Early postnatal phenomena thus pave the way for the immune system's adaptation to the external environment. For instance, immunosuppressive erythroid cells are abundant in early life [58]neutrophil numbers decline after birth [59]neonatal T cells exhibit a more innate-like behavior without establishing robust memory populations [60]and high expression levels of IL-10 and IL-27 are observed in neonates [59]. Furthermore, the developing microbiota plays a crucial role in shaping the adaptive immune response during this period. Clostridia, which become established at weaning, are primary producers of SCFAs that promote the differentiation of RORγt regulatory T cells. These RORγt^{+} Tregs modulate Th2 responses in the intestine and help regulate immune interactions with the microbiota throughout life [61, 62]. Additionally, Bifidobacterium longum subsp. infantis colonizes the infant intestine and produces indole-3-lactic acid, which

Table 3 Microbial dominance dynamics in host colonization post-HSCT

Microbial Contestant	Dominance Pattern	Competitive Advantage	Host Response	GVHD Implications	Ref.
Enterococcus	Frequent post-HSCT domination	Exploitation of ecological vacuum; Antibiotic resistance	Exacerbation of mucosal damage; Systemic inflammation	Increased GVHD severity	[50]
Clostridiales species	Beneficial colonizers from donor	SCFA production; Metabolic versatility	Reinforcement of intestinal epithelial integrity; Suppression of pro-inflammatory T cell responses	Reduced GVHD severity	[5]
Lactobacillus	Competitive with beneficial commensals	Niche occupation in disrupted gut environments	Variable host responses	GVHD exacerbation via MHC-II-dependent activation of alloreactive T cells	[51]
Bifidobacterium	Often displaced post-HSCT	Limited competitive advantage in disrupted environment	Loss associated with reduced immune regulation	Loss correlates with increased inflammatory complications	
Residual recipient Enterococcus	Exploits post-transplant niche	Adaptation to disrupted gut ecology	Increased inflammation and mucosal damage	Exacerbation of GVHD severity	[50]

Abbreviations: GVHD: graft-versus-host disease; HSC: hematopoietic stem cell; HSCT: hematopoietic stem cell transplantation; MHC-II: major histocompatibility complex class II; SCFA: Short-chain fatty acid upregulates the immunomodulatory molecule galectin-1 in $CD4^+$ T cells, thereby reducing their differentiation into Th2 and Th17 phenotypes [63]. Other specific Bifidobacterium species similarly suppress monocyte-derived inflammatory cytokines via indole acid production and enhance gut barrier integrity through AhR activation [64].

Interactions between the immune system and the microbiome are profoundly reciprocal, with each shaping the other's function. This dynamic crosstalk establishes critical pathways by which both endogenous and exogenous factors modulate systemic physiology [65]. Exogenous influences, notably dietary components, can perturb these interactions and thereby impact various organs. For instance, a diet enriched in saturated fats fosters the proliferation of *Bilophila wadsworthia* via the accumulation of taurocholic acid—a secondary bile acid that promotes Th1-type immune responses and predisposes IL-$10^{-/-}$ deficient mice to colitis [66]. Conversely, intermittent fasting enriches the gut microbiota with members of the Lactobacillaceae, Bacteroidaceae, and Prevotellaceae families while enhancing antioxidative microbial metabolic pathways that decrease IL-17–producing T cells and promote regulatory T cell induction, ultimately ameliorating experimental autoimmune encephalomyelitis (EAE) and multiple sclerosis (MS) [67]. Moreover, secondary bile acids generated from primary bile acids in the colon act as critical endogenous regulators of gut $ROR\gamma t^+$ regulatory T cell homeostasis [68].

Microbial metabolites and immune modulation

Microbiota do not typically interact directly with epithelial cells through attachment—as seen in the gut, where mucosal layers provide a protective barrier—but rather influence both local and systemic physiology. The primary mediators of these effects are microbiota-derived products, including metabolites, short-chain fatty acids, long-chain fatty acids, and various antigens [65]. For instance, polysaccharide A from the commensal *Bacteroides fragilis* is recognized by a Toll-like receptor (TLR)1/2 heterodimer in conjunction with Dectin-1. This interaction activates downstream phosphoinositide 3-kinase (PI3K) signaling, leading to the inactivation of glycogen synthase kinase 3β (GSK3β) and subsequent induction of cAMP response element binding protein (CREB)-dependent anti-inflammatory gene expression [69, 70]. Moreover, the gut microbiome finely tunes immune system function, culminating in widespread systemic changes. For example, segmented filamentous bacteria (SFB) promote non-inflammatory Th17 cell differentiation, whereas certain Citrobacter-derived components drive an inflammatory Th17 phenotype [71].

Colonic and intestinal bacteria ferment dietary fibers to produce SCFAs, key metabolites that impact both local and systemic cellular functions. SCFAs are fatty acids containing 1–6 carbon atoms, notably acetic acid (acetate), propionic acid (propionate), and butyric acid (butyrate) [72]. Butyrate is predominantly absorbed by the gut epithelium, propionate by the liver, and acetate by peripheral tissues [31]. Once absorbed, these metabolites modulate intracellular signaling by binding to G protein-coupled receptors (GPCRs), including GPR41, GPR43, and GPR109A. These receptors are expressed across diverse tissues and display variable affinities for individual SCFAs, thereby mediating distinct physiological functions. For instance, GPR41 is ubiquitously expressed, whereas GPR43 is primarily confined to lymphoid tissues and immune cells; both bind acetate, propionate, and butyrate, although GPR109A predominantly recognizes butyrate [72, 73]. Engagement of these receptors represses cAMP-dependent signaling pathways while activating alternative cascades such as mTOR and pSTAT3 [74]. Consequently, SCFA-induced signaling promotes activation of the NLRP3 inflammasome and IL-18 release by colonic epithelial cells, and enhances the differentiation and regulatory functions of $FOXP3^+$ T cells [73].

SCFAs enter cells either by passive diffusion or via specialized transporters, including the sodium-coupled monocarboxylate transporter 1 (SMCT1, encoded by *SLC5A8*) and the monocarboxylate transporter 1 (MCT1, encoded by *SLC16A1*), exerting concentration-dependent effects. At low concentrations, butyrate and propionate are metabolized to acetyl-CoA, which serves as an acetyl donor for histone acetyltransferases such as p300, thereby enhancing histone acetylation [75]. Intracellularly, butyrate and propionate also inhibit HDAC activity, further promoting an open chromatin state that modulates gene expression. This epigenetic reprogramming influences diverse cellular processes—including differentiation, cell cycle regulation, oxidative stress responses, inflammation, antimicrobial defense, and fatty acid metabolism—in macrophages, dendritic cells, lymphocytes (particularly regulatory T cells), and intestinal epithelial cells [76–79]. Moreover, additional epigenetic modifications, such as propionylation, butyrylation, and crotonylation mediated by butyrate, can alter transcription factor binding to chromatin [80, 81]. Recent studies have demonstrated that butyrate's HDAC inhibitory activity modulates lysine acetylation of the NF-κB subunit p65 via HDAC3 and HDAC6, thereby influencing the recruitment of NF-κB to pro-inflammatory gene promoters both in vitro and in vivo (Fig. 1) [82].

Microbiome disruption and GVHD development

The intestinal microbiota exerts a profound influence on the development and severity of acute GVHD by shaping the host's immune environment even before

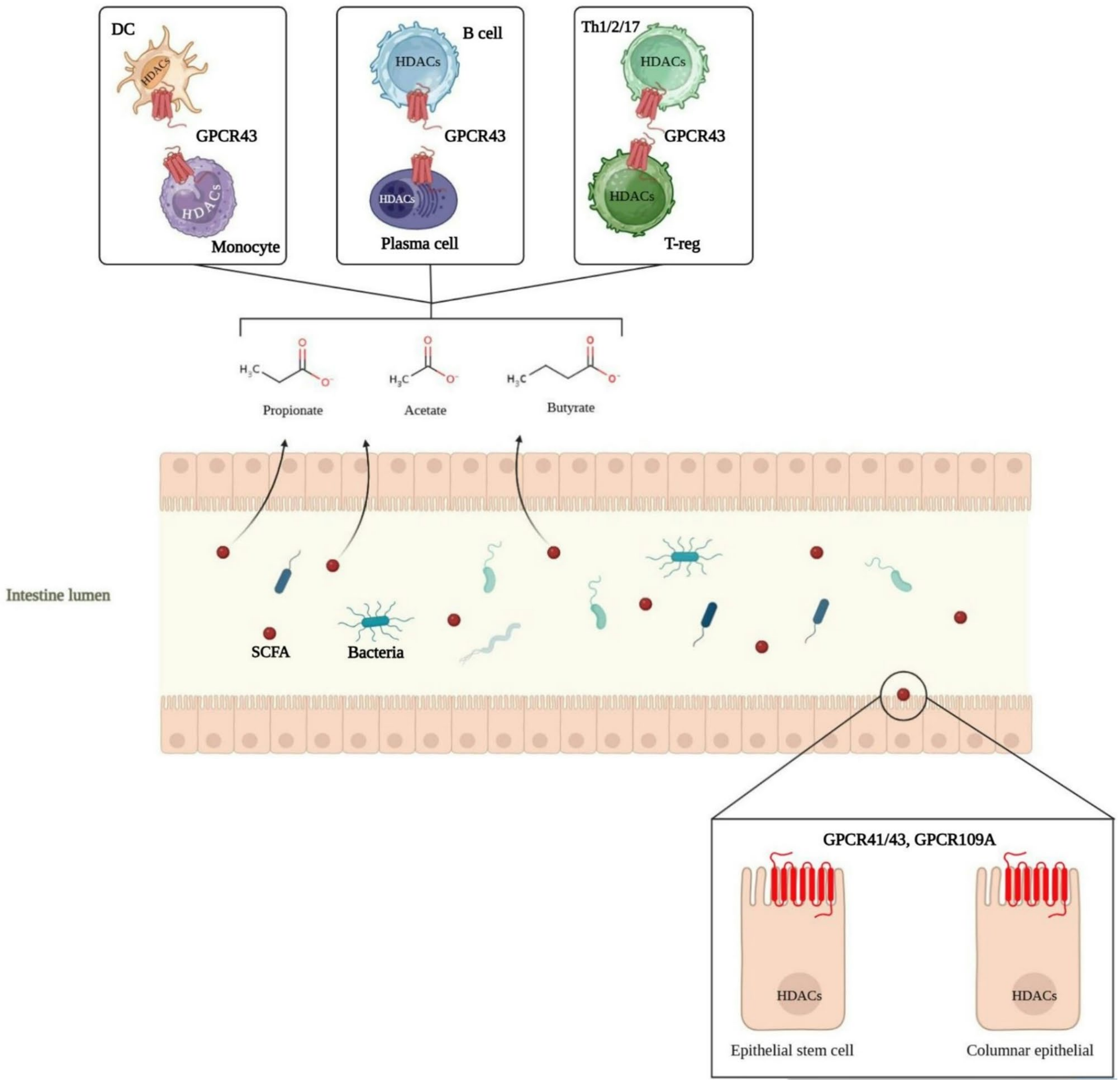

Fig. 1 SCFAs regulate immune responses in the gut through GPCR signaling and HDACs modulation. SCFAs produced by gut bacteria are absorbed by intestinal epithelial cells via GPCR41/43 and GPCR109A, affecting epithelial stem cells and columnar epithelial cells. These metabolites shape immune homeostasis by interacting with GPCR43 and HDACs on dendritic cells (DCs), monocytes, B cells, plasma cells, Th1/2/17 cells, and regulatory T cells, contributing to immune modulation in the gut

transplantation. Conditioning therapies—including chemotherapy and radiotherapy—cause significant mucosal injury and inflammation that compromise the intestinal barrier, thereby allowing bacterial translocation [83, 84]. This disruption not only triggers local immune activation but also leads to a loss of microbial diversity and a reduction in key metabolites such as SCFAs, which are essential for barrier repair and anti-inflammatory signaling. Reduced microbial diversity has been correlated with increased transplant-related and GVHD-related mortality, underscoring the critical role of a healthy microbiome in mitigating GVHD severity [85].

Recent investigations have revealed that the composition of the intestinal microbiota directly modulates immune responses through effects on antigen presentation by intestinal epithelial cells (IECs). Studies have shown that specific bacterial taxa can act as MHC-II inducers or suppressors, thereby influencing the expression of MHC-II molecules on IECs [86]. Enhanced MHC-II expression—driven by microbiota-induced local production of interferon-gamma (IFN-γ)—promotes the activation of donor-derived $CD4^+$ T cells, which are key mediators of GVHD. Conversely, targeted depletion of MHC-II inducer bacteria via selective antibiotic regimens, such as oral vancomycin, leads to a reduction in MHC-II expression and subsequent attenuation of GVHD severity [87].

Beyond these direct immunological effects, the overall balance of microbial communities and the abundance of specific bacterial genera further dictate GVHD outcomes. A progressive decline in microbial diversity—initiated during remission-inducing chemotherapy and exacerbated post-transplant—results in the overgrowth of pathobionts such as Enterococcus and Streptococcus, both of which are potent stimulators of intestinal inflammation. In contrast, beneficial taxa like *Barnesiella* and *Lactobacillus johnsonii* help maintain microbial homeostasis by suppressing the expansion of harmful bacteria [24]. Moreover, diminished levels of SCFAs in both the gut and circulation have been linked to increased disease severity, highlighting the therapeutic potential of restoring a balanced microbiota and its metabolic outputs in allogeneic HCT recipients [88].

Dysbiosis—an imbalance in the gut microbial community—can compromise the integrity of the intestinal barrier, allowing microbial products such as lipopolysaccharide (LPS) and other PAMPs to translocate into the circulation. This translocation activates innate immune cells via pattern recognition receptors (e.g., Toll-like receptors), which in turn release pro-inflammatory cytokines (TNF-α, IL-1β, IL-6) and chemokines (IL-8), thereby amplifying systemic inflammation—a central hallmark in the pathogenesis of GVHD [85]. Furthermore, dysbiosis may disturb the balance between regulatory and effector T cells. For instance, a reduction in Treg cells along with an increase in Th1 cells leads to elevated IFN-γ levels and a concomitant decrease in anti-inflammatory cytokines like IL-10. This skewed cytokine environment further fuels systemic inflammation in GVHD [89]. These findings underscore that strategic modulation of the intestinal microbiota and its metabolites holds significant promise in attenuating donor T cell activation, preserving gut barrier integrity, and ultimately mitigating acute GVHD severity while enhancing overall transplant outcomes (Table 4).

Gut-Liver Axis and immune regulation

The gut–liver axis constitutes a complex immunometabolic network in which microbial-derived metabolites, particularly SCFAs such as butyrate, play a pivotal role. Butyrate, produced in the colon through fermentation of dietary fibers, is delivered to the liver via the portal circulation and can contribute up to 30% of the hepatic energy supply. Once in the liver, butyrate exerts anti-inflammatory effects by acting on Kupffer cells to enhance the production of immunosuppressive prostaglandin E_2 (PGE_2) and concurrently suppress pro-inflammatory cytokines like TNF-α [90].

Mechanistically, butyrate modulates transcriptional regulation by increasing H3K9 acetylation at the promoter region of peroxisome proliferator-activated receptor alpha (PPARα), thereby attenuating NF-κB-mediated inflammatory signaling. These findings underscore butyrate's role as both a metabolic substrate and an epigenetic regulator, contributing to hepatic homeostasis and resilience against inflammatory insults [72, 91].

Disruption of this axis, as seen in early-life microbial dysbiosis—often precipitated by antibiotic exposure—can have profound downstream effects on hepatic immunity, including the functional maturation of liver-resident natural killer (LrNK) cells [92]. Recent studies demonstrate that early antibiotic treatment reduces hepatic butyrate levels, impairing IL-18 production in Kupffer cells and hepatocytes via a GPR109A-dependent mechanism. The resultant deficiency in IL-18 signaling undermines mitochondrial function and arrests the maturation of LrNK cells, thereby compromising their capacity for immunosurveillance and tolerance. In the context of GVHD, where aberrant donor T cell activation drives systemic inflammation and liver injury, impaired LrNK cell function may exacerbate hepatic damage. Thus, strategies aimed at restoring butyrate levels—for instance, through dietary supplementation or administration of butyrogenic probiotics—could potentially re-establish

Table 4 Microbial factors and their impact on GVHD. Summarizes how different microbial components influence GVHD severity, their mechanisms, and potential therapeutic strategies

Microbial Factor	Effect on GVHD	Mechanism	Therapeutic Potential
Microbial Diversity	High diversity reduces severity; low increases mortality	Maintains gut barrier, immune homeostasis	FMT to restore diversity
Beneficial Bacteria (e.g., *Lactobacillus*, *Barnesiella*)	Suppress progression	Inhibit pathobionts, enhance Tregs	Probiotics with these strains
Pathobionts (e.g., *Enterococcus*, *Streptococcus*)	Exacerbate severity	Trigger inflammation via LPS translocation	Antibiotics (e.g., vancomycin)
SCFAs (e.g., Butyrate)	Reduced levels worsen; supplementation mitigates	Repair gut barrier, induce Tregs, reduce inflammation	Prebiotics to boost SCFA production
MHC-II Inducing Bacteria	Enhance severity	Increase IEC MHC-II, activate donor T cells	Selective antibiotic depletion

IL-18-mediated NK cell maturation and mitigate liver pathology in GVHD patients [93].

Therapeutic opportunities

GVHD is a major cause of morbidity and mortality post-allo-HSCT, characterized by donor immune cells attacking recipient tissues, with the gastrointestinal tract being a primary target. Current treatments, primarily corticosteroids, are effective in only 40–60% of patients, with many experiencing steroid dependency or resistance [94]. The gut microbiota's role in GVHD pathogenesis has spurred interest in microbiota-based interventions. This review evaluates probiotic intervention, prebiotic supplementation, and FMT, focusing on their potential to prevent and treat GVHD, drawing from recent studies and ongoing trials.

Probiotics

Probiotics, defined as live microorganisms conferring health benefits, have been explored for their potential to mitigate GVHD by restoring gut microbiome balance and enhancing barrier function. A pilot randomized clinical trial by Yazdandoust et al. (2023) suggested that synbiotic intake (probiotics plus prebiotics) before and during conditioning may reduce acute GVHD incidence and severity by inducing CD4 + CD25 + Foxp3 + regulatory T cells, potentially improving transplant outcomes [95]. However, this study was small ($n = 30$), limiting generalizability.

Safety was assessed in pediatric populations by Ladas et al. (2016), finding *Lactobacillus plantarum* administration feasible, with 97% compliance and no bacteremia in 30 children and adolescents undergoing HCT, though efficacy in GVHD prevention was not evaluated [96]. Animal studies, such as Sofi et al. (2021), showed *Bacteroides fragilis* improving gut barrier integrity and reducing GVHD in mice by enhancing mucus production and reducing epithelial damage, suggesting strain-specific benefits [97].

Further animal studies have demonstrated the potential of probiotics in GVHD management. For instance, oral administration of Lactobacillus Rhamnosus GG before and after transplantation resulted in improved survival and reduced aGVHD in murine models [98]. Similarly, combining Lactobacillus acidophilus with the immunosuppressant FK506 reduced inflammation and the severity of GVHD compared to FK506 alone in both in vivo and in vitro studies. These findings underscore the potential of specific probiotic strains in modulating immune responses and enhancing gut barrier function to mitigate GVHD [99].

However, concerns regarding the safety of administering live microorganisms to immunocompromised patients persist. While some probiotics have been deemed safe during aGVHD, such as Lactobacillus plantarum in pediatric patients and Lactobacillus Rhamnosus GG in murine models, there have been reports of sepsis, bacteremia, and meningitis following probiotic treatment in immunocompromised individuals. These incidents highlight the necessity for cautious selection and monitoring of probiotic use in vulnerable populations [100].

Ongoing clinical trials aim to further elucidate the role of probiotics in GVHD prevention. A randomized phase III trial is currently investigating the efficacy of Lactobacillus plantarum in preventing aGVHD in children undergoing donor stem cell transplantations. The outcomes of such studies are anticipated to provide more definitive evidence regarding the safety and effectiveness of probiotics in this context(NCT03057054).

Prebiotics

Prebiotics, non-digestible fibers promoting beneficial bacteria growth, are investigated for their role in maintaining gut microbiome homeostasis. Holmes et al. used Galacto-Oligosaccharides (GOS) in mice, finding it prevented antibiotic-induced gut injury post-allo-HCT by expanding butyrogenic bacteria and increasing SCFA production, potentially modulating alloreactivity [101]. Another review, noted prebiotics like GOS reduced GVHD severity in murine models, but cautioned that lactose may enhance GVHD by reducing microbiota diversity [102].

Clinical studies have also explored the role of prebiotics in GVHD prevention. A phase I/II trial is currently assessing the safety, optimal dosing, and efficacy of GOS in preventing aGVHD in patients undergoing allogeneic HSCT. This study aims to determine whether GOS supplementation can modulate the gut microbiome and reduce the incidence of aGVHD, thereby improving transplant outcomes (NCT04373057).

Furthermore, the timing of prebiotic administration appears to be critical. Prebiotics are most effective in individuals with a relatively intact and diverse microbiome, as they serve as substrates for existing beneficial bacteria. In patients with low microbial diversity prior to transplantation, the benefits of prebiotic supplementation may be diminished. Therefore, assessing the baseline gut microbiome composition could inform the strategic timing and potential efficacy of prebiotic interventions in the context of HSCT [103].

Fecal microbiota transplantation (FMT)

FMT, which involves transferring fecal matter from healthy donors to patients, has emerged as a promising approach to restore gut microbiota balance in individuals with GVHD, particularly those with steroid-refractory manifestations. A comprehensive meta-analysis by Li et al. (2022) pooled data from 79 patients across six studies

Table 5 Microbiota-Based therapeutic interventions for GVHD. Outlines probiotics, prebiotics, FMT, and personalized therapies, highlighting their benefits, supporting evidence, and challenges

Intervention	Potential Benefits	Key Evidence	Challenges
Probiotics	Restores gut balance, enhances barrier function, reduces aGVHD incidence/severity	- Synbiotics induced Tregs, reduced aGVHD - *L. plantarum* safe in kids - *B. fragilis*, *L. rhamnosus GG* reduced GVHD in mice.	Safety risks (e.g., sepsis), inconsistent outcomes, strain-specific effects, colonization issues.
Prebiotics	Promotes beneficial bacteria, increases SCFAs, reduces GVHD severity	- GOS prevented gut injury, boosted SCFAs in mice - GOS reduced GVHD in models.	Depends on intact microbiome, potential adverse effects (e.g., lactose reducing diversity).
Fecal Microbiota Transplantation (FMT)	Restores diversity, effective for steroid-refractory GVHD, reduces aGVHD incidence	– 82.4% response rate in meta-analysis - Ameliorated intestinal GVHD - Prophylactic FMT safe, effective	Recurrence, mortality risks, donor selection, standardization, long-term safety concerns.
Personalized Therapies	Predicts GVHD risk, tailors interventions for better outcomes	- Pre-transplant microbiome profiles predict GVHD risk	Requires further validation, complexity in profiling and implementation.

and five case reports, revealing a complete remission (CR) rate of 55.9% and a partial remission (PR) rate of 26.5%, culminating in an overall response rate of 82.4%. Importantly, the treatment was associated with low mortality and predominantly mild adverse effects, underscoring its potential efficacy and safety profile [104]. Individual studies have further substantiated these findings. For instance, Weber et al. reported that donor FMT ameliorated intestinal GVHD in allogeneic hematopoietic cell transplant recipients, highlighting its therapeutic potential [105].

However, challenges persist, including the recurrence of symptoms and mortality from causes such as hepatic GVHD and septic shock. These underscore the necessity for careful patient selection and monitoring during FMT therapy. Notably, a randomized, double-blind, placebo-controlled trial published in Nature Communications (2025) evaluated the prophylactic use of FMT early after alloHCT to prevent acute GVHD. The study demonstrated that FMT was feasible and safe, with potential benefits in reducing GVHD incidence, thereby suggesting a role for FMT not only as a therapeutic intervention but also as a preventive strategy [47].

Collectively, emerging evidence suggests that interventions targeting the gut microbiota—whether through probiotics, prebiotics, or fecal microbiota transplantation—hold considerable promise for mitigating graft-versus-host disease. Probiotics may enhance intestinal barrier function and modulate immune responses, while prebiotics support the growth of beneficial bacteria and the production of short-chain fatty acids, both of which contribute to a healthier microbial environment. Fecal microbiota transplantation, in particular, has demonstrated high response rates in refractory cases by restoring microbial diversity and promoting gastrointestinal healing. Despite these encouraging findings from preclinical studies, pilot trials, and meta-analyses, further large-scale, controlled clinical investigations are essential to determine optimal dosing strategies, ensure long-term safety, and establish standardized treatment protocols that can be reliably integrated into clinical practice.

Challenges and future directions

Despite promising preliminary data, several challenges and limitations remain for microbiota-targeting strategies in GVHD prevention and treatment. Probiotic approaches face significant hurdles including strain-specific efficacy, difficulties in ensuring adequate colonization, and safety concerns in immunocompromised patients. For instance, while certain strains have shown the ability to modulate immune responses and enhance barrier function in preclinical models, clinical trials have yielded inconsistent outcomes and raised issues such as the risk of bacteremia and low survival of ingested organisms in the harsh gastrointestinal environment. Moreover, the complexity of the gut ecosystem means that simply adding a few strains may not restore a balanced microbiome in the setting of profound dysbiosis.

Prebiotics similarly depend on the presence of a relatively intact microbial community to stimulate the growth of beneficial bacteria and produce key metabolites like short-chain fatty acids. In patients with severely disrupted microbiota, prebiotics may have diminished effects or even adverse consequences—such as certain carbohydrates inadvertently reducing microbial diversity or exacerbating inflammation. FMT has demonstrated encouraging response rates, particularly in steroid-refractory GVHD, yet it is constrained by issues of donor selection, standardization of protocols, and long-term safety data. Heterogeneity among studies, potential transmission of pathogens, and regulatory uncertainties further complicate the clinical application of FMT (Table 5).

Personalized microbiome-based therapies for GVHD represent a promising frontier that leverages each patient's unique gut microbial signature to predict, prevent, and even treat graft-versus-host disease. Research has consistently shown that patients with reduced microbial diversity or an imbalanced community—characterized, for example, by low levels of beneficial taxa such as

Blautia and high levels of pro-inflammatory organisms like Enterobacteriaceae—are at increased risk for more severe GVHD and higher mortality following allo-HSCT [106, 107].

Moreover, recent predictive modeling work—using metagenome-wide association studies—has shown that pre-transplant gut microbiota profiles can achieve moderate predictive accuracy (AUC ~ 0.67) for GVHD risk, suggesting that such models could be integrated into clinical decision-making to customize prophylactic and therapeutic strategies [108]. Integrating high-resolution microbiome profiling into the transplant workflow opens the door for truly personalized microbiome-based interventions that could improve GVHD outcomes by restoring a balanced gut ecosystem, tailoring dietary or probiotic strategies, and even selecting donors with a favorable microbial profile. This precision approach not only has the potential to reduce the incidence and severity of GVHD but also to enhance overall survival in transplant patients.

Global disparities in access to microbiota-based treatments, such as probiotics, prebiotics, and FMT, significantly impact the management of GVHD in allogeneic HSCT settings. While FMT has demonstrated an 82.4% response rate in steroid-refractory GVHD cases [104]and clinical trials for probiotics (e.g., Lactobacillus plantarum, NCT03057054) and prebiotics (e.g., galacto-oligosaccharides, NCT04373057) show promise, these interventions are predominantly available in high-resource regions due to high costs, specialized infrastructure requirements, and regulatory barriers. In low- and middle-income countries, patients often rely on conventional immunosuppressive therapies, which achieve only a 40–60% response rate [94]exacerbating inequities in transplant outcomes. Addressing these disparities requires the development of cost-effective, scalable solutions, such as locally produced probiotics or simplified FMT protocols, alongside international efforts to standardize treatment guidelines and enhance access to microbiome-targeted therapies, thereby improving GVHD management globally.

Conclusion

The intricate interplay between the microbiome and immune system profoundly influences GVHD pathogenesis, offering novel therapeutic opportunities to reshape transplant outcomes. Dysbiosis in HSCT recipients, driven by chemotherapy, radiation, and antibiotics, disrupts microbial metabolites critical for immune regulation, barrier integrity, and inflammatory control. Conversely, restoring microbial diversity through probiotics, prebiotics, or FMT has shown potential to attenuate GVHD by rebalancing T-cell subsets, enhancing SCFA production, and curbing systemic inflammation. Despite promising preclinical and clinical data, challenges persist—including safety concerns, variable efficacy of microbial interventions, and the need for standardized protocols. Future research must prioritize large-scale trials to validate these approaches, refine predictive models using microbiome profiling, and develop personalized therapies tailored to individual microbial and immune landscapes. By bridging microbial ecology with immunology, this evolving field holds transformative potential to mitigate GVHD severity, reduce non-relapse mortality, and enhance the success of allogeneic HSCT. The microbiome, once a neglected player, now stands at the forefront of precision medicine in transplantation.

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Author contributions
MHM, MA were involved in the conception of the study, data analysis, manuscript preparation, and monitoring. Moreover, SM, MM, MA, and HSS contributed to the search for relevant manuscripts and the preparation of the manuscript. MA, SM, and MHM contributed to the development of the search strategy, article search, and manuscript preparation. Finally, all the authors reviewed and approved the manuscript.

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Data availability
The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

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